

CS 791 - Assignment 1 Report

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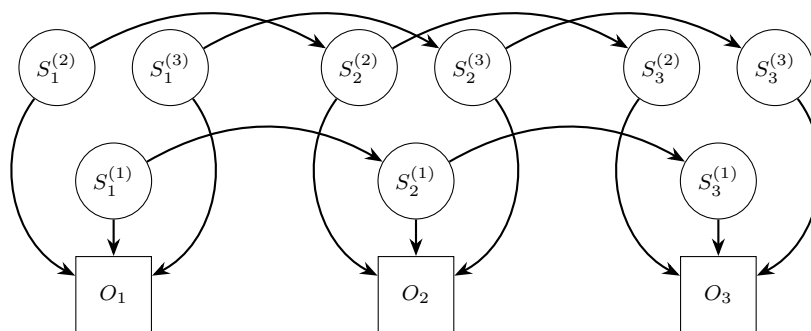
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1 Factorial Hidden Markov Model (FHMM)

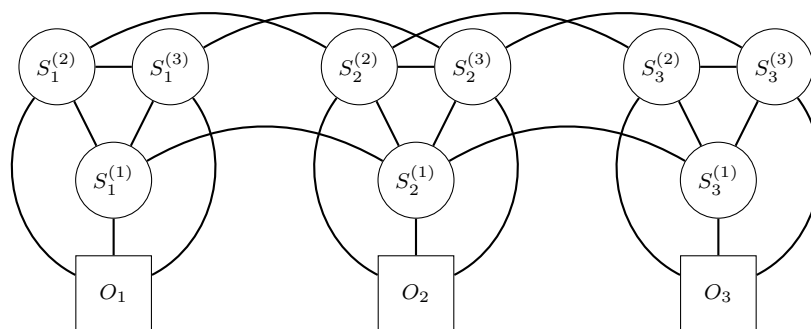
An FHMM is an extension of the standard Hidden Markov Model (HMM) that allows for multiple independent hidden state sequences to influence the observed data.

The problem of interest is to infer the marginals of the hidden states, and the most probable sequence of hidden states, given a sequence of observations in a Factorial Hidden Markov Model (FHMM).

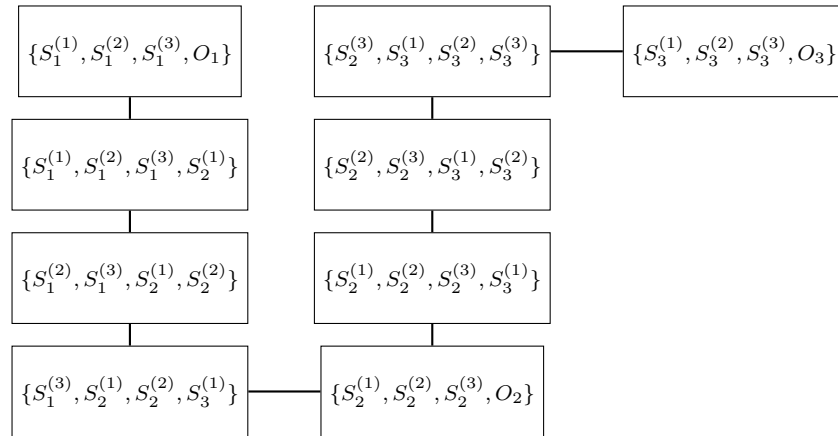
The problem involved making a junction tree out of the FHMM by triangulating it and then using the message passing algorithm to infer the marginals of the hidden states and the most probable sequence of hidden states given a sequence of observations.



The moralized graph looks like



The junction tree made out of the cliques after triangulation is a chain as shown below:



The message passing algorithm is now simple with just two passes, one from left to right and the other from right to left.

For including the prior of the observations, we merged the observations into the cliques by removing the observation nodes and considering the potentials over the cliques that include the observations.

For finding the z-value of the junction tree, the message passing algorithm runs over every clique for